

Second Entry to the Generalized t -SNE Problem

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1 Current State of the Problem

For the big picture, we aim to establish the convergence theorems for the t -SNE algorithm with generalized input and output kernels. For convenience, if not specified or otherwise explicitly defined, all notations are the same as in [Auffinger and Fletcher, 2023]. At the currently state, we have shown that the relative entropy t -SNE with generalized kernel g converges and can be computed if g satisfies the following:

$g(y, y') \equiv k(\|y - y'\|)$ such that:

1. $k : \mathbb{R}^+ \rightarrow \mathbb{R}^+$ is decreasing and bounded.
2. g satisfies

$$\iint_{(0, \infty)^2} g(y, y') dy dy' < \infty. \quad (1.1)$$

3. k has bounded derivative.
4. $k'(0) = 0$.

More importantly, the good news is that the proofs related to the generalized output kernel does not interfere with those of input kernel. This significantly reduces the complexity and potential roadblocks for us to continue working on the generalized input kernel.

There are two main steps before we can complete the project. On one hand, we need to give a specific form to the generalized input kernel, and prove the existence and uniqueness of σ^* . Without the constraint on perplexity, t -SNE will fail. On the other hand, we need to prove the bounds for approximation and produce similar results like those in [Auffinger and Fletcher, 2023]. This document is dedicated to the former, and serves as the second main step in this project, hence the name.

2 The Generalized Input Kernel

The most important question that sets the foundation of the problem is, what does the output kernel look like, given two data points x, x' and parameter σ_i . We can start by assuming that the input kernel only depends on the distance between inputs, i.e., $\|x - x'\|$ because t -SNE is not about the absolute coordinates, but rather the relations between data points. Then, we may further assume that the input kernel is a function of $\sigma \|x - x'\|^\theta$ for some fixed $\theta > 0$, as σ is used to control the scaling of distance between two data points in the original algorithm. This way we can control the entropy of the induced probability distributions, and it also aligns with the very reason we introduce the concept of perplexity.

Finally, because the input kernel is about the probability mass and should always be positive, we can write it as $\exp(\cdot)$ as the exponential function is a bijection. In particular, we write the input kernel as

$$\exp(-w(\sigma_i \|x_j - x_i\|^\theta)), \quad (2.1)$$

where $w : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}_{\geq 0}$ is a monotone increasing function and $\theta > 0$ is a fixed constant. In turn, the induced probability mass will be

$$p_{j|i} = \frac{\exp(-w(\sigma_i \|x_j - x_i\|^\theta))}{\sum_{k \neq i} \exp(-w(\sigma_i \|x_k - x_i\|^\theta))}, \quad (2.2)$$

Naturally, the continuous version of the kernel would be

$$f_\psi(x, x') = \exp(-w(\psi(x) \|x - x'\|^\theta)), \quad (2.3)$$

for some function $\psi : \mathbb{R}^d \rightarrow \mathbb{R}^+$. Such expression has several advantages:

1. Since the exponential function is a bijection, we do not lose much generality at all except for how σ interact with the output.
2. The form is in direct comparison with the original kernel. As we dive deeper into the computation we can have clearer insights about the constraints we need for the input kernel.
3. The exponential function will cancel out the logarithm in the formula of entropy, greatly simplifying the computation.
4. The form looks natural and aligns with the original concept and motivation of entropy and perplexity.

With the above being said, we can start adding assumptions to the input kernel. The most immediate and obvious requirements we can impose will be

$$\lim_{t \rightarrow \infty} w(t) = \infty, \quad \int_0^\infty \exp(-w(t^\theta)) dt < \infty. \quad (2.4)$$

These two are the bare minimum for this problem because apparently we want the normalizing quantity to be finite and w must increase to infinity for a decaying tail. Taking $w(t) = t/2$, $\theta = 2$ and we have the original kernel back. We will start with these basic settings and gradually add more requirements as we proceed through the problem. Also, since the input kernel serves as the weight/probability mass, it is always normalized and we can assume $w(0) = 0$ without loss of generality.

3 Existence and Uniqueness of σ^*

We will try to replicate the results in section 2 of [Auffinger and Fletcher, 2023]. Recall that

$$\begin{aligned} & F_{\rho, \mu}(x, \sigma) \\ &= \int \frac{\exp(-w(\sigma \|x - x'\|^\theta))}{\int \exp(-w(\sigma \|x - x'\|^\theta)) d\mu(x')} \log \left(\frac{\exp(-w(\sigma \|x - x'\|^\theta))}{\int \exp(-w(\sigma \|x - x'\|^\theta)) d\mu(x')} \right) d\mu(x') + \log \rho \\ &= - \int \frac{w(\sigma \|x - x'\|^\theta) \exp(-w(\sigma \|x - x'\|^\theta))}{\int \exp(-w(\sigma \|x - x'\|^\theta)) d\mu(x')} d\mu(x') \\ &\quad - \log \left(\int \exp(-w(\sigma \|x - x'\|^\theta)) d\mu(x') \right) + \log \rho. \end{aligned} \quad (3.1)$$

We want to show that $F_{\rho,\mu}(x,\sigma) = 0$ has a unique solution $\sigma^* = \sigma_{\rho,\mu}^*(x)$ for all $\rho \in (0,1)$ and $x \in \mathbb{R}^d$. First off, we have $\lim_{\sigma \rightarrow 0} F_{\rho,\mu}(x,\sigma) = \log \rho < 0$. We will then divide the estimation into two steps. In the first step, we will prove that $F_{\rho,\mu}(x,\sigma)$ has a lower bound that increases to ∞ as we let $\sigma \rightarrow \infty$. In the second step, we explore and prove the monotonicity of $F_{\rho,\mu}(x,\sigma)$. Combine both steps and we have the existence and uniqueness of σ^* by intermediate value theorem.

Lower Bound

Let x be in the support of μ so we have $f(x) > 0$. In this section we aim to establish a lower bound of $F_{\rho,\mu}(x,\sigma)$ that goes to infinity as we increase σ . Assume μ has C^1 density f and both $f, \|\nabla f\|$ are bounded. Recall

$$\int_0^\infty \exp(-w(t^\theta)) dt < \infty. \quad (3.2)$$

For any integrable function h on $\mathbb{R}_{\geq 0}$, we have

$$\int_{\mathbb{R}^d} h(\|x\|) dx = \int_{\mathbb{S}^{d-1}} \int_0^\infty r^{d-1} h(r) dr d\nu = S_{d-1} \int_0^\infty r^{d-1} h(r) dr, \quad (3.3)$$

where $\mathbb{S}^{d-1}$ is the $(d-1)$ -dimensional sphere, $d\nu$ the corresponding surface measure, and S_{d-1} the volume of the sphere. Therefore it is necessary that we assume

$$\int_0^\infty t^{d-1} \exp(-w(t^\theta)) dt < \infty. \quad (3.4)$$

Denote

$$Z_d := \int_{\mathbb{R}^d} \exp(-w(\|\tau\|^\theta)) d\tau = S_{d-1} \int_0^\infty t^{d-1} \exp(-w(t^\theta)) dt. \quad (3.5)$$

Consider

$$\begin{aligned} \int \exp(-w(\sigma \|x - x'\|^\theta)) d\mu(x') &= \int \exp(-w(\sigma \|x - x'\|^\theta)) f(x') dx' \\ &= Z_d \sigma^{-d/\theta} \int \frac{\exp(-w(\sigma \|x - x'\|^\theta))}{Z_d \sigma^{-d/\theta}} f(x') dx'. \end{aligned} \quad (3.6)$$

Because $\lim_{x \rightarrow \infty} w(x) = \infty$, the integrand of the above integral converges weakly to $\delta_x f(x')$. To prove this statement, notice that

$$\begin{aligned} &\int \frac{\exp(-w(\sigma \|x - x'\|^\theta))}{Z_d \sigma^{-d/\theta}} f(x') dx' \\ &= \frac{1}{Z_d} \int \exp\left(-w\left(\left\|\sigma^{1/\theta}(x - x')\right\|^\theta\right)\right) f(x') d\sigma^{1/\theta}(x - x') \\ &\equiv \frac{1}{Z_d} \int \exp(-w(\|\tau\|^\theta)) f(x - \tau \sigma^{-1/\theta}) d\tau. \end{aligned} \quad (3.7)$$

Since f is bounded, $\exp(-w(\|\tau\|^\theta)) \sup f$ is integrable. By dominant convergence theorem and the continuity of f , we have

$$\begin{aligned} &\lim_{\sigma \rightarrow \infty} \frac{1}{Z_d} \int \exp(-w(\|\tau\|^\theta)) f(x - \tau \sigma^{-1/\theta}) d\tau \\ &= \frac{1}{Z_d} \int \lim_{\sigma \rightarrow \infty} \exp(-w(\|\tau\|^\theta)) f(x - \tau \sigma^{-1/\theta}) d\tau \\ &= f(x) \frac{1}{Z_d} \int \exp(-w(\|\tau\|^\theta)) d\tau \\ &= f(x). \end{aligned} \quad (3.8)$$

There exists some $\epsilon(\sigma) \rightarrow 0$ as $\sigma \rightarrow \infty$ and a constant $C > 0$, that

$$\begin{aligned} -\log \left(\int \exp(-w(\sigma \|x - x'\|^\theta)) d\mu(x') \right) &= -\log \left(Z_d \sigma^{-d/\theta} \right) - \log f(x) + \epsilon(\sigma) \\ &\geq C \log \sigma - \log f(x). \end{aligned} \quad (3.9)$$

Next, since f is bounded, there exists $M > 0$ such that $f \leq M$.

$$\begin{aligned} &\int \frac{w(\sigma \|x - x'\|^\theta) \exp(-w(\sigma \|x - x'\|^\theta))}{\int \exp(-w(\sigma \|x - x'\|^\theta)) d\mu(x')} d\mu(x') \\ &\leq M \frac{\int w(\sigma \|x - x'\|^\theta) \exp(-w(\sigma \|x - x'\|^\theta)) dx'}{\int \exp(-w(\sigma \|x - x'\|^\theta)) f(x') dx'} \\ &= M \frac{\sigma^{-d/\theta} \int w(\|\sigma^{1/\theta}(x - x')\|^\theta) \exp(-w(\|\sigma^{1/\theta}(x - x')\|^\theta)) d\sigma^{1/\theta}(x - x')}{\int \exp(-w(\sigma \|x - x'\|^\theta)) f(x') dx'} \\ &= M \frac{\sigma^{-d/\theta} \int w(\|\tau\|^\theta) \exp(-w(\|\tau\|^\theta)) d\tau}{\int \exp(-w(\sigma \|x - x'\|^\theta)) f(x') dx'} \\ &\rightarrow \frac{M}{Z_d f(x)} \int w(\|\tau\|^\theta) \exp(-w(\|\tau\|^\theta)) d\tau < \infty \end{aligned} \quad (3.10)$$

if we assume

$$\int_0^\infty t^{d-1} w(t^\theta) \exp(-w(t^\theta)) dt < \infty. \quad (3.11)$$

As the bound in (3.10) does not depend on σ , we can combine (3.1), (3.9) and (3.10) and deduce that

$$F_{\rho, \mu}(x, \sigma) \geq C \log \sigma - \log f(x) + \log \rho. \quad (3.12)$$

Monotonicity

To prove monotonicity, we take derivative of $F_{\rho, \mu}(x, \sigma)$ with respect to σ using (3.1). The following simplified notations will be used in the first several steps:

$$w = w(\sigma \|x - x'\|^\theta), \quad w' = w'(\sigma \|x - x'\|^\theta), \quad \exp(\cdot) = \exp(-w(\sigma \|x - x'\|^\theta)). \quad (3.13)$$

Omit $d\mu(x')$ in the integration for cleaner writing, and we have

$$\begin{aligned} &\frac{\partial}{\partial \sigma} F_{\rho, \mu}(x, \sigma) \\ &= - \int \frac{w' \|x - x'\|^\theta \exp(\cdot)}{\int \exp(\cdot)} - \int \frac{w \exp(\cdot)'}{\int \exp(\cdot)} + \int \frac{w \exp(\cdot) (\int \exp(\cdot))'}{(\int \exp(\cdot))^2} - \frac{(\int \exp(\cdot))'}{\int \exp(\cdot)} \\ &= - \int \frac{w \exp(\cdot)'}{\int \exp(\cdot)} + \int \frac{w \exp(\cdot) (\int \exp(\cdot))'}{(\int \exp(\cdot))^2} \quad (\text{The first and fourth terms cancel.}) \\ &= \int \frac{w \cdot w' \cdot \|x - x'\|^\theta \cdot \exp(\cdot)}{\int \exp(\cdot)} - \frac{\int w \cdot \exp(\cdot) \int w' \cdot \|x - x'\|^\theta \exp(\cdot)}{(\int \exp(\cdot))^2}. \end{aligned} \quad (3.14)$$

We are able to move differentiation inside the integral because of dominant convergence theorem. To show that $\frac{\partial}{\partial \sigma} F_{\rho, \mu}(x, \sigma) \geq 0$, we just need to show

$$\int w \cdot w' \cdot \|x - x'\|^\theta \cdot \exp(\cdot) \int \exp(\cdot) \geq \int w \cdot \exp(\cdot) \int w' \cdot \|x - x'\|^\theta \exp(\cdot). \quad (3.15)$$

To tackle this inequality, we need the following lemma:

Lemma 1 (Theorem 7, [Liu et al., 2009]). Let $f(x), g(x), h(x)$ be continuous functions on $[a, b]$ such that

$$(g(x) - g(y)) \left(\frac{f(y)}{h(y)} - \frac{f(x)}{h(x)} \right) \geq 0. \quad (3.16)$$

Then the inequality

$$\frac{\int_a^b f(x) dx}{\int_a^b h(x) dx} \geq \frac{\int_a^b f(x) g(x) dx}{\int_a^b h(x) g(x) dx} \quad (3.17)$$

holds. If (3.16) reverses, then (3.17) reverses.

This lemma can easily be extended to general Lebesgue integrals over $\mathbb{R}^d$ if every integral converges, which is our case.

Lemma 2. Let $f(x), g(x), h(x)$ be continuous functions $\mathbb{R}^d \rightarrow \mathbb{R}^+$ such that for all $x, y \in \mathbb{R}^d$

$$(g(x) - g(y)) \left(\frac{f(y)}{h(y)} - \frac{f(x)}{h(x)} \right) \geq 0. \quad (3.18)$$

Let μ be an arbitrary Lebesgue measure and assume all integrals in this lemma converge. Then the inequality

$$\frac{\int_{\mathbb{R}^d} f(x) d\mu}{\int_{\mathbb{R}^d} h(x) d\mu} \geq \frac{\int_{\mathbb{R}^d} f(x) g(x) d\mu}{\int_{\mathbb{R}^d} h(x) g(x) d\mu} \quad (3.19)$$

holds. If (3.18) reverses, then (3.19) reverses.

Proof. The proof of this lemma is similar to the one in [Liu et al., 2009]. It suffices to show that

$$\int_{\mathbb{R}^d} f(x) d\mu \int_{\mathbb{R}^d} h(x) g(x) d\mu \geq \int_{\mathbb{R}^d} h(x) d\mu \int_{\mathbb{R}^d} f(x) g(x) d\mu. \quad (3.20)$$

The above inequality is equivalent to

$$\int_{\mathbb{R}^d} f(x) d\mu(x) \int_{\mathbb{R}^d} h(y) g(y) d\mu(y) \geq \int_{\mathbb{R}^d} h(x) d\mu(x) \int_{\mathbb{R}^d} f(y) g(y) d\mu(y). \quad (3.21)$$

That means we need to show

$$\mathcal{G} := \int_{\mathbb{R}^d} \int_{\mathbb{R}^d} g(y) (f(x) h(y) - f(y) h(x)) d\mu(x) d\mu(y) \geq 0. \quad (3.22)$$

Swap x and y in the formula of $\mathcal{G}$ and we have an equivalent expression:

$$\mathcal{G} = \int_{\mathbb{R}^d} \int_{\mathbb{R}^d} g(x) (f(y) h(x) - f(x) h(y)) d\mu(x) d\mu(y). \quad (3.23)$$

Adding the last two formulas together yields

$$\begin{aligned} 2\mathcal{G} &= \int_{\mathbb{R}^d} \int_{\mathbb{R}^d} (g(x) - g(y)) (f(y) h(x) - f(x) h(y)) d\mu(x) d\mu(y) \\ &= \int_{\mathbb{R}^d} \int_{\mathbb{R}^d} h(x) h(y) (g(x) - g(y)) \left(\frac{f(y)}{h(y)} - \frac{f(x)}{h(x)} \right) d\mu(x) d\mu(y) \geq 0 \end{aligned} \quad (3.24)$$

by the hypothesis of the lemma. This finishes the proof. $\square$

For each $x, y, x' \in \mathbb{R}$ if the equality

$$(w'(\sigma \|x - x'\|^\theta) \|x - x'\|^\theta - w'(\sigma \|y - x'\|^\theta) \|y - x'\|^\theta) \cdot \left(\frac{1}{w(\sigma \|y - x'\|^\theta)} - \frac{1}{w(\sigma \|x - x'\|^\theta)} \right) \geq 0 \quad (3.25)$$

holds, then by letting $f(x) = \exp(\cdot)$, $h(x) = w \cdot \exp(\cdot)$ and $g(x) = w' \cdot \|x - x'\|^\theta$, we have (3.15) to be true by lemma 2. Let us take a closer look at the condition. Let $\|x - x'\|^\theta > \|y - x'\|^\theta$ without loss of generality. Then by monotonicity of w , we have $w(\sigma \|x - x'\|^\theta) > w(\sigma \|y - x'\|^\theta)$. In order for (3.25) to be true, we want

$$w'(\sigma \|x - x'\|^\theta) \|x - x'\|^\theta - w'(\sigma \|y - x'\|^\theta) \|y - x'\|^\theta \geq 0. \quad (3.26)$$

Since $\sigma > 0$, the above inequality is equivalent to

$$w'(\sigma \|x - x'\|^\theta) \sigma \|x - x'\|^\theta - w'(\sigma \|y - x'\|^\theta) \sigma \|y - x'\|^\theta \geq 0. \quad (3.27)$$

Let $\tilde{w}(x) = xw'(x)$. Based on (3.27) and mean value theorem, $\tilde{w}$ should satisfy for all $a, b \geq 0$, $a > b$, there exists $c \in [b, a]$ that

$$\frac{a\tilde{w}(a) - b\tilde{w}(b)}{a - b} = \tilde{w}'(c) = w'(c) + cw''(c) \geq 0. \quad (3.28)$$

Because a, b are arbitrary, the most generalized condition we can impose on w is for any $x \geq 0$,

$$w'(x) + xw''(x) \geq 0. \quad (3.29)$$

On the other hand, restricting w to be convex and (3.27) will obviously be true. In comparison, (3.29) is a less concise condition. We will see if we can carry this condition through the next phase of the paper. If the proofs for subsequent lemmas become too complicated because of condition in (3.29), we can always restrict w further to be convex.

Wrap Up

We know that $\lim_{\sigma \rightarrow 0} F_{\rho, \mu}(x, \sigma) = \log \rho < 0$ and by (3.12) $\lim_{\sigma \rightarrow \infty} F_{\rho, \mu}(x, \sigma) = \infty$. Moreover, since $F_{\rho, \mu}(x, \sigma)$ is monotone increasing in σ , we can conclude there exists a unique $\sigma^* = \sigma_{\rho, \mu}^*(x)$ for all $\rho \in (0, 1)$ and $x \in \mathbb{R}^d$.

4 Conclusion

In this document we have given a specific form to the general kernel. We also proved the existence and uniqueness of $\sigma_{\rho, \mu}^*(x)$. Let us summarize:

The generalized t -SNE has the following kernel

$$p_{j|i} = \frac{\exp(-w(\sigma_i \|x_j - x_i\|^\theta))}{\sum_{k \neq i} \exp(-w(\sigma_i \|x_k - x_i\|^\theta))}. \quad (4.1)$$

Here σ_i is used to control entropy (thus perplexity) and $\theta > 0$ is a fixed constant. The range of θ is remained to be determined in the next part of the problem. (We can replace σ_i with $\psi : \mathbb{R}^d \rightarrow \mathbb{R}^+$ in the continuous version of the kernel.) $w : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}_{\geq 0}$ is a function that satisfies:

1. $w'(t) > 0$ for all $t \geq 0$ with $\lim_{t \rightarrow \infty} w(t) = \infty$;
2. $w'(t) + xw''(t) \geq 0$ for all $t \geq 0$;
3. The following two integrals converge:

$$\int_0^\infty t^{d-1} \exp(-w(t^\theta)) dt < \infty, \quad \int_0^\infty t^{d-1} w(t^\theta) \exp(-w(t^\theta)) dt < \infty. \quad (4.2)$$

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